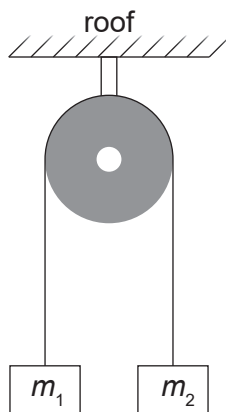


PAPER 1 / QUESTION 49

28 April 2022 / TZ1 / HL Paper 1 / original question 6 / 2222-6507

49. Two masses m_1 and m_2 are connected by a string over a frictionless pulley of negligible mass. The masses are released from rest. Air resistance is negligible.



Mass m_2 accelerates downwards at $\frac{g}{2}$. What is $\frac{m_1}{m_2}$?

- A. $\frac{1}{3}$
- B. $\frac{1}{2}$
- C. 2
- D. 3